

# WAVEFRONTS FOR DIFFUSION-CONVECTION REACTION EQUATIONS

Diego Berti

Dipartimento di Scienze e Metodi dell'Ingegneria  
Università di Modena e Reggio Emilia

**UNIMORE**  
UNIVERSITÀ DEGLI STUDI DI  
MODENA E REGGIO EMILIA



Seminari di Analisi @ Unife  
Ferrara (online), April 29th 2021

# DIFFUSION-CONVECTION REACTION EQUATIONS

We deal with the following equation:

$$\boxed{\rho_t + f(\rho)_x = \{D(\rho) \rho_x\}_x + g(\rho) \text{ for } x \in \mathbb{R}, t \geq 0.} \quad (1)$$

Here, the unknown is  $\rho \in [0, 1]$  (which may be understood as a **density of pedestrians, vehicles, population,...**) while the given “coefficients” of (1) are supposed to satisfy, in order:

(f)  $f \in C^1[0, 1]$  with  $f(0) = 0$  and denote  $h = \frac{d}{d\rho} f$ ; (**convection**)

–  $D \in C^1[0, 1]$  which vanishes in  $[0, 1]$ ; (**degenerate diffusion**)

–  $g \in C^0[0, 1]$  with  $g > 0$  in  $(0, 1)$ ; (**reaction**)

(g0)  $g(0) = 0 < g(1)$ ;

(g01)  $g(0) = g(1) = 0$  ( $g$  is of monostable-type).

# DIFFUSION-CONVECTION REACTION EQUATIONS

We deal with the following equation:

$$\rho_t + f(\rho)_x = \{D(\rho) \rho_x\}_x + g(\rho) \quad \text{for } x \in \mathbb{R}, t \geq 0. \quad (1)$$

Here, the unknown is  $\rho \in [0, 1]$  (which may be understood as a **density of pedestrians, vehicles, population,...**) while the given “coefficients” of (1) are supposed to satisfy, in order:

(f)  $f \in C^1[0, 1]$  with  $f(0) = 0$  and denote  $h = \frac{d}{d\rho} f$ ; (**convection**)

–  $D \in C^1[0, 1]$  which vanishes in  $[0, 1]$ ; (**degenerate diffusion**)

–  $g \in C^0[0, 1]$  with  $g > 0$  in  $(0, 1)$ ; (**reaction**)

(g0)  $g(0) = 0 < g(1)$ ;

(g01)  $g(0) = g(1) = 0$  ( $g$  is of monostable-type).

The zeros of  $g$  are the stationary solutions of (1).

Solutions of (1) are determined up to spatial shifts.

## SOME MOTIVATIONS: COLLECTIVE MOVEMENTS

There are several motivations to study parabolic equations as (1). About **collective movements (vehicular flows and crowd dynamics)**:

- for general information, [Garavello-Han-Piccoli, 2016], [Garavello-Piccoli, 2006], [Rosini, 2013];
- for degenerate parabolic equations, [Bellomo-Delitala-Coscia, MMMAS 2002], [Bellomo-Dogbe, SIAM R 2021], [Bruno-Tosin-Tricerri-Venuti, AMM 2011];
- for sign-changing diffusivities, [Corli-Malaguti, ZAMP 2019], [Corli-Malaguti, Lecture Notes INDAM, 2020(?)], [Nelson, PRE 2000].

Roughly speaking, in this modeling  $\rho$  represents the normalized density of the agents and  $f(\rho) = \rho v(\rho)$  the corresponding flux, where  $v$  is the prescribed velocity. Diffusion is included to avoid unrealistic shock waves in the equation.

## SOME MOTIVATIONS: COLLECTIVE MOVEMENTS

There are several motivations to study parabolic equations as (1). About **collective movements (vehicular flows and crowd dynamics)**:

- for general information, [Garavello-Han-Piccoli, 2016], [Garavello-Piccoli, 2006], [Rosini, 2013];
- for degenerate parabolic equations, [Bellomo-Delitala-Coscia, MMMAS 2002], [Bellomo-Dogbe, SIAM R 2021], [Bruno-Tosin-Tricerri-Venuti, AMM 2011];
- for sign-changing diffusivities, [Corli-Malaguti, ZAMP 2019], [Corli-Malaguti, Lecture Notes INDAM, 2020(?)], [Nelson, PRE 2000].

Roughly speaking, in this modeling  $\rho$  represents the normalized density of the agents and  $f(\rho) = \rho v(\rho)$  the corresponding flux, where  $v$  is the prescribed velocity. Diffusion is included to avoid unrealistic shock waves in the equation.

In this framework:  $D < 0$  in some zones models aggregation phenomena due to limited visibility conditions or panic behaviours. A monostable term  $g$  models entries that are forbidden when the density is either 0 (aggregation phenomena) or 1 (physical impossibility of enterings).

## TRAVELING-WAVE SOLUTIONS

*Traveling wave* solutions (TWs for short) are solutions  $\rho$  which takes the form

$\rho(x, t) = \varphi(x - ct)$ , for some wave profile  $\varphi = \varphi(\xi)$  and a wave speed  $c \in \mathbb{R}$ . Under this assumption, we formally rephrase (1) as

$$\left( D(\varphi) \varphi' \right)' + (c - h(\varphi)) \varphi' + g(\varphi) = 0. \quad (\text{ode})$$

We are interested on TWs whose profile is **non-constant** and **monotone (decreasing)**. As a consequence, we look for profiles satisfying

$$\lim_{\xi \rightarrow -\infty} \varphi(\xi) = 1 \quad \text{and} \quad \lim_{\xi \rightarrow +\infty} \varphi(\xi) = 0, \quad (2)$$

if  $g(0) = 0 = g(1)$ , or simply,

$$\lim_{\xi \rightarrow +\infty} \varphi(\xi) = 0, \quad (3)$$

if  $g(0) = 0$  and  $g(1) > 0$ .

## TRAVELING-WAVE SOLUTIONS

*Traveling wave* solutions (TWs for short) are solutions  $\rho$  which takes the form

$\rho(x, t) = \varphi(x - ct)$ , for some wave profile  $\varphi = \varphi(\xi)$  and a wave speed  $c \in \mathbb{R}$ . Under this assumption, we formally rephrase (1) as

$$(D(\varphi)\varphi')' + (c - h(\varphi))\varphi' + g(\varphi) = 0. \quad (\text{ode})$$

We are interested on TWs whose profile is **non-constant** and **monotone (decreasing)**. As a consequence, we look for profiles satisfying

$$\lim_{\xi \rightarrow -\infty} \varphi(\xi) = 1 \quad \text{and} \quad \lim_{\xi \rightarrow +\infty} \varphi(\xi) = 0, \quad (2)$$

if  $g(0) = 0 = g(1)$ , or simply,

$$\lim_{\xi \rightarrow +\infty} \varphi(\xi) = 0, \quad (3)$$

if  $g(0) = 0$  and  $g(1) > 0$ .

**Example 1 (Fisher equation).** The Fisher equation is defined as  $u = u(t, x) \in (0, 1)$  satisfying

$$u_t = \sigma u_{xx} + \lambda u(1 - u).$$

for some  $\sigma, \lambda > 0$ . Here,  $D(u) = \sigma$ ,  $g(u) = \lambda u(1 - u)$  and  $f = 0$ .

Traveling waves from 1 to 0 exist **if and only if**  $c \geq 2\sqrt{\lambda\sigma}$ .

## SIMPLE EXAMPLES

**Example 1 (Fisher equation).** The Fisher equation is defined as  $u = u(t, x) \in (0, 1)$  satisfying

$$u_t = \sigma u_{xx} + \lambda u(1 - u).$$

for some  $\sigma, \lambda > 0$ . Traveling waves from 1 to 0 exist **if and only if**  $c \geq 2\sqrt{\lambda\sigma}$ .

**Example 2 (Allen-Cahn reaction term).** Consider the following equation

$$\rho_t = \rho_{xx} + \rho - \rho^3, \quad \text{for } \rho \in [-1, 1],$$

so that  $D(\rho) = 1$ ,  $g(\rho) = \rho - \rho^3$  which **changes sign** and  $f = 0$ . Here the corresponding elliptic equation

$$-u_{xx} = u - u^3$$

is solvable in  $\mathbb{R}$  by

$$u(x) = \tanh(x/\sqrt{2}),$$

so that the **unique** (up to space shifts) traveling wave from 1 to 0 is given by

$$\rho(t, x) = u(x), \quad x \in \mathbb{R} \quad \text{and} \quad t \geq 0.$$

## WHAT IS CURRENTLY KNOWN ABOUT TWs FOR (1) UNDER POSITIVE $g$ (VERY BRIEF LIST)

The analysis of TWs of (1) under sub-cases of assumptions above places among a consolidated thread of research. A (very, very brief) list of references is the following:

- [Aronson-Weinberger, *Adv. in Math.* 1978], where first it was tackled  $D = 1$  and  $f = 0$ ;
- [Marcelli-Malaguti, *Adv. Nonlinear Equations*, 2003] (case  $D \geq 0$ ,  $(g01)$  and general  $f$ ); [Gilding-Kersner, 2004] (for several results mainly in the case  $D \geq 0$ ,  $(g01)$  and  $f = 0$ );
- [Corli-di Ruvo-Malaguti-Rosini, *Netw. Heterog. Media* 2017], [Corli-Malaguti, *Dyn. Partial Differ. Equ.* 2016] (The case  $g(1) = 0 < g(0)$ .)
- [Bao-Zhao, *Discrete Contin. Dyn. Syst. Ser. B* 2014] - [Maini-Malaguti-Marcelli-Matucci, *Discrete Contin. Dyn. Syst. Ser. B* 2006] (for a changing-sign  $D$  and  $f = 0$ .)

### TYPICAL RESULT

There exists a **critical threshold**  $c^* \in \mathbb{R}$ , depending on both  $f$  and the product  $Dg$  such that

TWs of (1) satisfying (2) exist  $\iff c \geq c^*$ .

## OUR CONTRIBUTION

In collaboration with Corli (Ferrara) and Malaguti (Modena - Reggio Emilia), we have pushed further on the study about the existence and regularity of wavefronts (and semi-wavefronts) for Equation (1).

Our results are contained in the following papers:

- \* [B.-Corli-Malaguti, *Electron. J. Qual. Theory Differ. Equ.* 2020]: (1) with positive diffusivity and (one or two-equilibria) reaction ;
- \* [B.-Corli-Malaguti, (submitted to) *Discrete Contin. Dyn. Syst.* 2021]: (1) with sign-changing diffusivity and monostable reaction ;
- \* [B.-Corli-Malaguti, in preparation]: (1) with sign-changing diffusivity and sign-changing reaction .

# POSITIVE DEGENERATE DIFFUSIVITY

Assume:

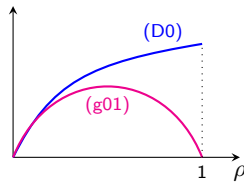
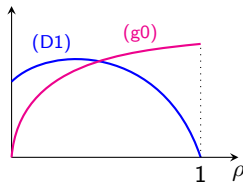
$$(D1) \quad D > 0 \text{ in } (0, 1), \quad D(1) = 0;$$

$$(g0) \quad g > 0 \text{ in } (0, 1], \quad g(0) = 0;$$

or

$$(D0) \quad D > 0 \text{ in } (0, 1), \quad D(0) = 0;$$

$$(g01) \quad g > 0 \text{ in } (0, 1), \quad g(0) = g(1) = 0.$$



## SOME FORMAL DEFINITIONS

We define now (formally) traveling waves and the related profiles. We denote with  $I \subseteq \mathbb{R}$  an open interval.

### DEFINITION OF TWS

Let  $c$  be a real constant and  $\varphi$  be a function such that

- $\varphi \in C(I)$  with values in  $[0, 1]$ , for some  $I \subset \mathbb{R}$ ;
- $\varphi$  is differentiable a.e.;
- $D(\varphi)\varphi' \in L^1_{\text{loc}}(I)$ .

The function  $\rho(x, t) := \varphi(x - ct)$ , for  $(x, t)$  with  $x - ct \in I$ , is a **traveling-wave** solution of equation (1) with wave speed  $c$  and wave profile  $\varphi$  if, for every  $\psi \in C_0^\infty(I)$ ,

$$\int_I (D(\varphi(\xi))\varphi'(\xi) - f(\varphi(\xi)) + c\varphi(\xi))\psi'(\xi) - g(\varphi(\xi))\psi(\xi) d\xi = 0.$$

A traveling-wave solution is

- **global** if  $I = \mathbb{R}$  and **strict** if  $I \neq \mathbb{R}$  and  $\varphi$  is not extendible to  $\mathbb{R}$ ;
- **classical** if  $\varphi$  is differentiable,  $D(\varphi)\varphi'$  is absolutely continuous and (ode) holds a.e.;  
**sharp at  $\ell$**  if there exists  $\xi_\ell \in I$  such that  $\varphi(\xi_\ell) = \ell$ , with  $\varphi$  classical in  $I \setminus \{\xi_\ell\}$  and not differentiable at  $\xi_\ell$ ;
- **a wavefront** if it is global, with a monotonic, non-constant profile  $\varphi$  satisfying either (2) or the converse condition;
- **a semi-wavefront to 1** (or *to 0*) if  $I = (a, \infty)$  for  $a \in \mathbb{R}$ , the profile  $\varphi$  is monotonic, non-constant and  $\varphi(\xi) \rightarrow 1$  (respectively,  $\varphi(\xi) \rightarrow 0$ ) as  $\xi \rightarrow \infty$ ; *a semi-wavefront from 1* (or *from 0*) if  $I = (-\infty, b)$  for  $b \in \mathbb{R}$ , the profile  $\varphi$  is monotonic, non-constant and  $\varphi(\xi) \rightarrow 1$  (respectively,  $\varphi(\xi) \rightarrow 0$ ) as  $\xi \rightarrow -\infty$ .

In [BCM1], we study semi-wavefronts and wavefront for (1), completing the analysis of previous works by Corli-Malaguti et al.

- There is a threshold  $c^*$  such that profiles only exist for  $c \geq c^*$ . We study their regularity and strict monotonicity.
- \* We give a refined estimate for  $c^*$ ;
- \* We improve a result obtained by Malaguti-Marcelli (under more tight assumptions on  $D$ ) about wavefronts with a sharp profile;
- \* In the case of semi-wavefronts we show that there exists a family of profiles with same speed  $c$ .

# WAVEFRONT AND SEMI-WAVEFRONT SOLUTIONS

- A TW is said **wavefront** if it is global ( $I = \mathbb{R}$ ), with a monotonic, non-constant profile  $\varphi$  satisfying (2).
- A TW is said **strict semi-wavefront to 0** of  $I = (\xi_1, +\infty)$ , with  $\xi_1 > -\infty$ , if the profile is monotonic, non-constant and (3) holds.

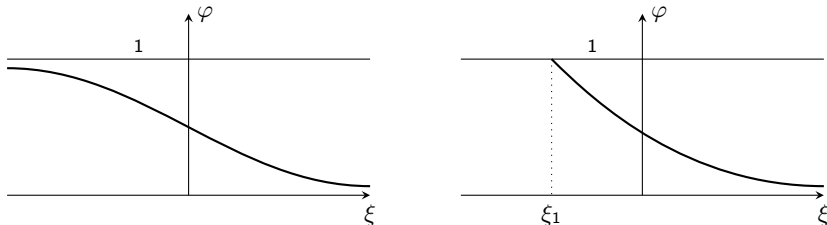


FIGURE: Left: a wavefront joining 1 with 0; right: a semi-wavefront to 0.

The main result about semi-wavefronts is the following:

### THEOREM 1 ([THEOREM 2.2, BCM1])

Assume  $(f)$ ,  $(D1)$ ,  $(g0)$  and

$$\limsup_{\varphi \rightarrow 0^+} \frac{D(\varphi)g(\varphi)}{\varphi} < +\infty. \quad (4)$$

Then, there exists  $c^* \in \mathbb{R}$ , which satisfies

$$\max \left\{ \sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi}, h(0) + 2\sqrt{\liminf_{\varphi \rightarrow 0^+} \frac{D(\varphi)g(\varphi)}{\varphi}} \right\} \leq c^* \leq 2\sqrt{\sup_{\varphi \in (0,1]} \frac{D(\varphi)g(\varphi)}{\varphi}} + \sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi},$$

such that (1) has strict semi-wavefronts to 0, connecting 1 to 0, if and only if  $c \geq c^*$ . Moreover, if  $\varphi$  is the profile of one of such semi-wavefronts, then it holds that

$$\varphi'(\xi) < 0 \quad \text{for any } 0 < \varphi(\xi) < 1.$$

## KEY TOOL: REDUCTION TO SINGULAR FIRST-ORDER PROBLEMS

A dynamical system analysis should require strong regularity assumptions on the coefficients ( $D$  and  $g$  and  $h$ ) and hence we need to investigate existence and regularity of fronts with a different method.

Our main tool is the **reduction** (in regions where  $D$  has constant sign) to singular first-order problems of the type (in case of **semi-wavefronts**)

$$\left\{ \begin{array}{ll} \dot{z}(\varphi) = h(\varphi) - c - \frac{D(\varphi)g(\varphi)}{z(\varphi)} & \varphi \in (0, 1), \\ z(\varphi) < 0 & \varphi \in (0, 1), \\ \textcolor{red}{z(0)} = 0. & \end{array} \right. \quad (5)$$

The above problem is deduced by (1)-(3) by the singular change of variable

$$z(\varphi) := D(\varphi)\varphi',$$

where the right-hand side is understood to be computed at  $\varphi^{-1}(\varphi)$ .

Equation (1) admits semi-wavefronts whose profile  $\varphi$  satisfies  $(??) \leftrightarrow$  there exists solutions of  $(??)$ .

## Comments on Theorem 1.

- For a fixed  $c > c^*$ , the **profiles of Theorem 1 are NOT unique**. This lack of uniqueness is due not only to the action of space shifts but, more intimately, to the non-uniqueness of solutions of

$$\begin{cases} \dot{z} = h - c - \frac{Dg}{z} & \text{in } (0, 1) \\ z < 0 & \text{in } (0, 1) \\ z(0) = 0. \end{cases}$$

given by [Proposition 5.1, BCM1]. The loss of uniqueness is a novelty compared with analogous results.

- There exists a  $\beta = \beta(c)$  such that the family of profiles can be written as

$$\varphi_b = \varphi_b(\xi), \quad b \in [\beta(c), 0].$$

Moreover,  $\beta(c) < 0$ , if  $c > c^*$  and  $\beta(+\infty) = -\infty$ .

- The threshold  $\beta(c)$  corresponds to the minimum value that  $D(\varphi_b)\varphi'_b$  achieves when  $\varphi_b$  reaches 1, for  $b \in [\beta(c), 0]$ .

On the estimates for  $c^*$ .

- The left estimate improves analogous bounds by including the term

$$\sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi} \geq h(0).$$

- If  $\dot{D}g(0) = 0$ , we obtain

$$\sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi} \leq c^* \leq 2 \sqrt{\sup_{\varphi \in (0,1]} \frac{Dg(\varphi)}{\varphi}} + \sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi},$$

which is significative since reveals clearly the separate contribution of the *convection component* and the *diffusion-reaction component*.

- This symmetry occurs in NONE of the previous estimates.

In some cases, wavefronts are sharp at 0 (i.e. they reaches 0 at finite value in a only continuous way).

The second main result in [BCM1]:

## THEOREM 2 ([THEOREM 2.3, BCM1])

Assume (f), (D0) and (g01) and (4). Then there exists  $c^*$ , satisfying

$$\sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi} \leq c^* \leq \sup_{\varphi \in (0,1]} \frac{f(\varphi)}{\varphi} + 2 \sqrt{\sup_{\varphi \in (0,1]} \frac{1}{\varphi} \int_0^\varphi \frac{D(\sigma)g(\sigma)}{\sigma} d\sigma},$$

such that Equation (1) admits a (unique) wavefront, whose profile  $\varphi$  satisfies (2),

if and only if  $c \geq c^*$ . We have  $\varphi'(\xi) < 0$ , for  $0 < \varphi(\xi) < 1$ , and

- if  $c > c^*$ , then  $\varphi$  is classical at 0;
- if  $c = c^*$  and  $c^* > h(0)$ , then  $\varphi$  is sharp at 0 and if it reaches 0 at  $\xi_0 \in \mathbb{R}$  then

$$\lim_{\xi \rightarrow \xi_0^-} \varphi'(\xi) = \frac{h(0) - c^*}{\dot{D}(0)} < 0.$$

Mainly, we obtained:

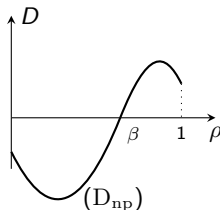
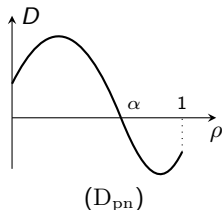
- New estimates on  $c^*$ ;
- New regularity classification which improves results in [MM2005] (under more tight assumptions on  $D$ ).

## SIGN-CHANGING DIFFUSIVITY

Assume that  $D$  changes sign once in  $(0, 1)$ , namely for some  $\alpha, \beta \in (0, 1)$ :

$(D_{pn})$   $D \in C^1[0, 1]$ ,  $D > 0$  in  $(0, \alpha)$  and  $D < 0$  in  $(\alpha, 1)$ ;

$(D_{np})$   $D \in C^1[0, 1]$ ,  $D < 0$  in  $(0, \beta)$  and  $D > 0$  in  $(\beta, 1)$ .



We deal with (1) when we assume  $(f)$ ,  $(g01)$  and  $(D_{pn})$  or  $(D_{np})$ .

The case of  $D$  changing sign twice or more (but finitely many) in  $(0, 1)$  are treated in [BCM2].

Case ( $D_{pn}$ ):  $D > 0$  in  $(0, \alpha)$  and  $D < 0$  in  $(\alpha, 1)$

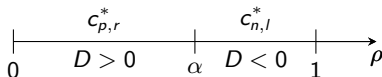
We also need to assume

$$\limsup_{\varphi \rightarrow 0^+} \frac{D(\varphi)g(\varphi)}{\varphi} < +\infty \quad \text{and} \quad \limsup_{\varphi \rightarrow 1^-} \frac{D(\varphi)g(\varphi)}{\varphi - 1} < +\infty. \quad (6)$$

We properly past together semi-wavefronts of the type given in [BCM1, Theorem 1]. Let  $c_{pn}^*$  be defined by

$$c_{pn}^* := \max \{c_{p,r}^*, c_{n,l}^*\},$$

where  $c_{p,r}^*$  and  $c_{n,l}^*$  are the thresholds for which semi-wavefronts connecting (resp.)  $\alpha$  to 0 and 1 to  $\alpha$  exist.



In order to generalize the case  $f = 0$  (mostly obtained in [MMMM2006]), we obtain the following results, where it is worth to emphasizing the appearance of **unusual** non-regular solutions.

### THEOREM 3 ([THEOREM 2.1, BCM2])

Assume (f),  $(D_{pn})$ ,  $(g01)$  and (4). Equation (1) admits a (unique up to space shifts) wavefront, with speed  $c$  and profile  $\varphi$  satisfying (2), if and only if  $c \geq c_{pn}^*$ .

For such  $c$ , we have  $\varphi'(\xi) < 0$  when  $\varphi(\xi) \in (0, 1) \setminus \{\alpha\}$ ; there exists a unique  $\xi_\alpha \in \mathbb{R}$  such that  $\varphi(\xi_\alpha) = \alpha$  and

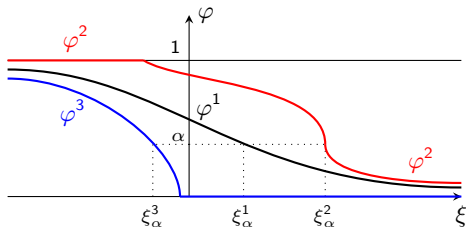
$$\varphi'(\xi_\alpha^\pm) = \begin{cases} \frac{g(\alpha)}{s_-(\alpha, c)} < 0 & \text{if } \dot{D}(\alpha) < 0 \text{ or } c > h(\alpha), \\ -\infty & \text{if } \dot{D}(\alpha) = 0 \text{ and } c \leq h(\alpha). \end{cases}$$

At last, it holds that:

- ❶ if  $D(0)D(1) < 0$ , then  $\varphi$  is strictly decreasing and hence classical in  $\mathbb{R} \setminus \{\xi_\alpha\}$ ;
- ❷ if  $D(0)D(1) = 0$  and  $c > c_{pn}^*$ , then  $\varphi$  is classical in  $\mathbb{R} \setminus \{\xi_\alpha\}$ ;
- ❸ if  $D(0) = 0$  and  $c = c_{pn}^* = c_{p,r}^* > h(0)$ , then  $\varphi$  is sharp at 0 (reached at  $\xi_0 \in \mathbb{R}$ ) and if  $D(1) = 0$  and  $c = c_{pn}^* = c_{n,l}^* > h(1)$ , then  $\varphi$  is sharp at 1 (reached at  $\xi_1 \in \mathbb{R}$ ). In these cases we have

$$\varphi'(\xi_0^-) = \frac{h(0) - c_{p,r}^*}{\dot{D}(0)} < 0 \text{ and } \varphi'(\xi_1^+) = \frac{h(1) - c_{n,l}^*}{\dot{D}(1)} < 0.$$

Some possible wavefronts joining 1 with 0 in case ( $D_{pn}$ ).



- a classical wavefront  $\varphi^1$  (for  $c > c_{pn}^*$  and either  $\dot{D}(\alpha) < 0$  or  $c > h(\alpha)$ );
- a wavefront  $\varphi^2$  which is sharp at 1, with finite right derivative at  $\xi_1^2$  and  $(\varphi^2)'(\xi_\alpha^2) = -\infty$  (for  $c = c_{n,l}^* > h(1)$ ,  $D(1) = 0$ ,  $\dot{D}(1) > 0$ ,  $\dot{D}(\alpha) = 0$  and  $c \leq h(\alpha)$ );
- a wavefront  $\varphi^3$  which is sharp at 0 with  $(\varphi^3)'(\xi_0^3) = -\infty$  (for  $c = c_{p,r}^* > h(0)$ ,  $D(0) = 0 = \dot{D}(0)$ ).

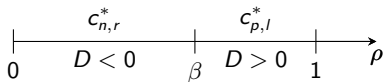
- For **every**  $c \geq c_{pn}^*$ , possibly  $(\varphi^2)'(\xi_\alpha^2) = -\infty$  occurs;
- If  $c > c_{pn}^*$  then the profile is classical at 0 and 1.

Case ( $D_{np}$ ):  $D < 0$  in  $(0, \beta)$  and  $D > 0$  in  $(\beta, 1)$

Let  $c_{np}^*$  be defined by

$$c_{np}^* := \max \{ c_{n,r}^*, c_{p,l}^* \},$$

where  $c_{n,r}^*$  and  $c_{p,l}^*$  are the thresholds for which semi-wavefronts connecting (resp.)  $\beta$  to 0 and 1 to  $\beta$  exist.



This result mostly generalizes [BZ2014], where the case  $f = 0$  were treated. The novelty is that **fronts only continuous** (at  $\xi_\beta$ ) are allowed.

## THEOREM 4 ([THEOREM 2.2, BCM2])

Assume (f),  $(D_{np})$  and (g01). Then, Equation (1) admits a (unique up to space shifts) wavefront, with speed  $c$  and profile  $\varphi$  satisfying (2), if and only if  $c \geq c_{np}^*$ .

For such  $c$ , we have  $\varphi' < 0$  if  $\varphi \in (0, 1) \setminus \{\beta\}$ . There exists a unique  $\xi_\beta \in \mathbb{R}$  such that  $\varphi(\xi_\beta) = \beta$  and

① if  $c > c_{np}^*$  we have

$$\varphi'(\xi_\beta^\pm) = \frac{g(\beta)}{s_-(\beta, c)};$$

② if  $c = c_{np}^* = c_{n,r}^* = c_{p,l}^*$  we have

$$\varphi'(\xi_\beta^\pm) = \frac{g(\beta)}{s_+(\beta, c_{np}^*)};$$

③ if  $c = c_{np}^* = c_{n,r}^* > c_{p,l}^*$  we have

$$\varphi'(\xi_\beta^-) = \frac{g(\beta)}{s_-(\beta, c_{np}^*)} > \varphi'(\xi_\beta^+) = \frac{g(\beta)}{s_+(\beta, c_{np}^*)};$$

④ if  $c = c_{np}^* = c_{p,l}^* > c_{n,r}^*$  we have

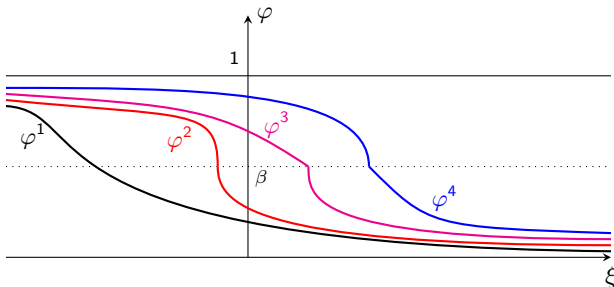
$$\varphi'(\xi_\beta^+) = \frac{g(\beta)}{s_-(\beta, c_{np}^*)} > \varphi'(\xi_\beta^-) = \frac{g(\beta)}{s_+(\beta, c_{np}^*)}.$$

[Theorem 2.2, BCM2] also contains the following **regularity results at 0 and 1**:

- ❶ if  $D(0)D(1) < 0$  then  $\varphi$  is **strictly decreasing and hence** classical in  $\mathbb{R} \setminus \{\xi_\beta\}$ ;
- ❷ if  $D(0) = 0$ , and either  $c > h(0)$  or  $c = h(0)$  and  $\dot{D}(0) < 0$ , then  $\varphi$  is classical at 0; if  $D(1) = 0$ , and either  $c > h(1)$  or  $c = h(1)$  and  $\dot{D}(1) < 0$  then  $\varphi$  is classical at 1;
- ❸ if  $D(0) = 0$  and  $c < h(0)$ , then  $\varphi$  is sharp at 0 (reached at some  $\xi_0 > \xi_\beta$ ); if  $D(1) = 0$  and  $c < h(1)$ , then  $\varphi$  is sharp at 1 (reached at some  $\xi_1 < \xi_\beta$ ). In these cases we have

$$\varphi'(\xi_0^-) = \frac{h(0) - c}{\dot{D}(0)} \quad \text{and} \quad \varphi'(\xi_1^+) = \frac{h(1) - c}{\dot{D}(1)}.$$

Some possible wavefronts joining 1 with 0 in case  $(D_{np})$ . For simplicity we only represented strictly monotone profiles.



- For  $c > c_{np}^*$   $\varphi$  is classical at  $\beta$  but it can be sharp at 0 or 1;
- For  $c = c_{np}^*$  the profile can be only continuous at  $\beta$ , 0 and 1.

Thanks for your attention!